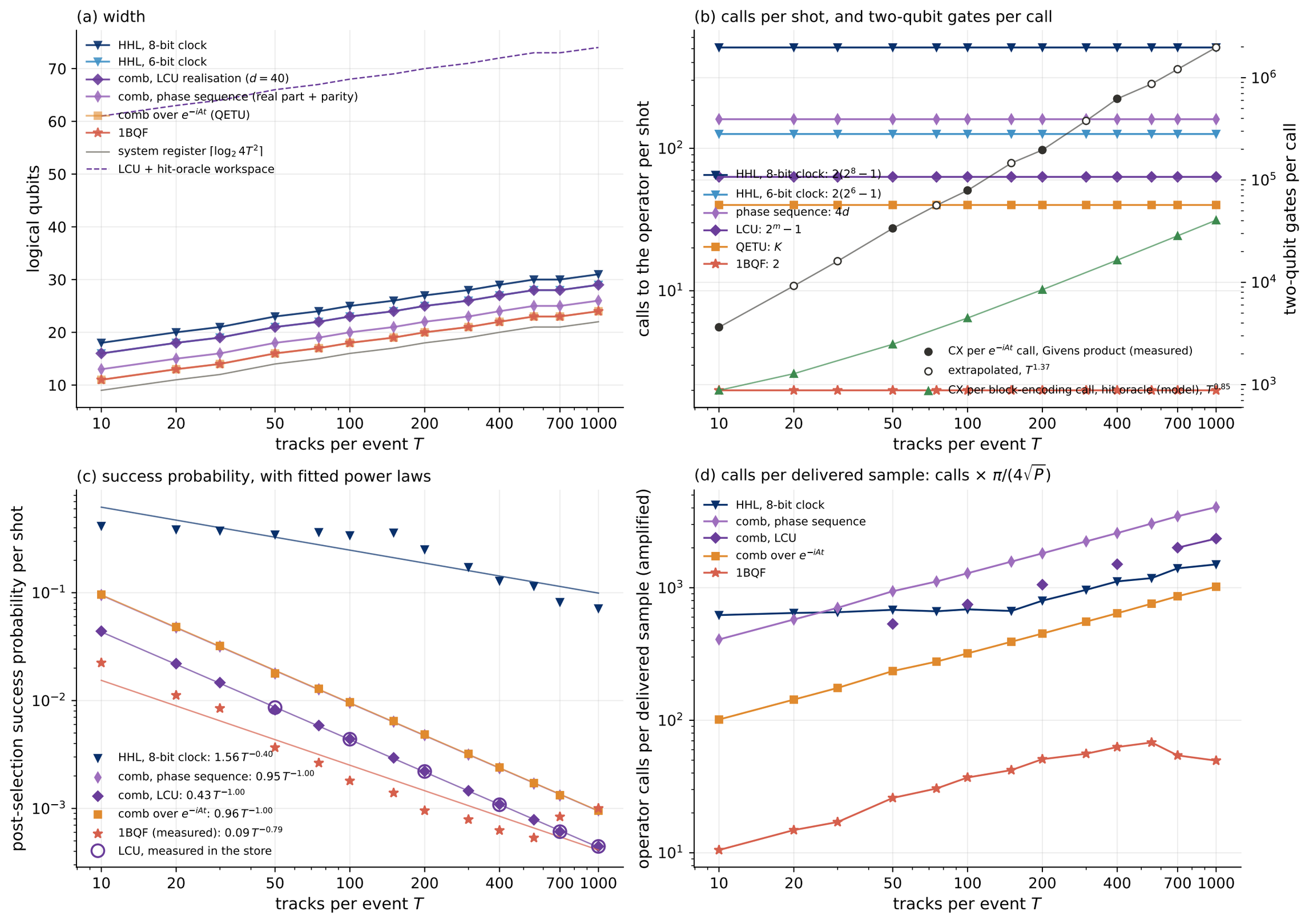
(a) width
HHL, 8-bit clock
HHL, 6-bit clock
comb, LCU realisation ($d = 40$)
comb, phase sequence (real part + parity)
comb over e^{-iAt} (QETU)
1BQF
system register $\lceil \log_2 4T^2 \rceil$
LCU + hit-oracle workspace
logical qubits
tracks per event T
(b) calls per shot, and two-qubit gates per call
HHL, 8-bit clock: $2(2^8 - 1)$
HHL, 6-bit clock: $2(2^6 - 1)$
phase sequence: $4d$
LCU: $2^m - 1$
QETU: K
1BQF: 2
CX per e^{-iAt} call, Givens product (measured)
extrapolated, $T^{1.37}$
CX per block-encoding call, hit oracle (model), $T^{0.85}$
calls to the operator per shot
two-qubit gates per call
tracks per event T
(c) success probability, with fitted power laws
HHL, 8-bit clock: $1.56\,T^{-0.40}$
comb, phase sequence: $0.95\,T^{-1.00}$
comb, LCU: $0.43\,T^{-1.00}$
comb over e^{-iAt}: $0.96\,T^{-1.00}$
1BQF (measured): $0.09\,T^{-0.79}$
LCU, measured in the store
post-selection success probability per shot
tracks per event T
(d) calls per delivered sample: calls × $\pi/(4\sqrt{P})$
HHL, 8-bit clock
comb, phase sequence
comb, LCU
comb over e^{-iAt}
1BQF
operator calls per delivered sample (amplified)
tracks per event T